

Python File Handling (Theory + Code) Backend & AI/ML Guide

1. Introduction

File handling allows Python programs to read, write, and manage files. It is essential in backend development (saving data, logs, uploads) and AI/ML (loading datasets, saving models).

2. Opening Files

Use `open(filename, mode)`. Modes define how the file is used.

Modes: r (read), w (write), a (append), x (create), b (binary), t (text).

```
file = open("example.txt", "r")
```

3. Reading Files

Read content from files using `read()`, `readline()`, or `readlines()`.

```
with open("example.txt", "r") as file:  
    print(file.read())
```

4. Writing Files

Write or overwrite data using `write()` in 'w' mode.

```
with open("example.txt", "w") as file:  
    file.write("Hello Krishna")
```

5. Appending Files

Append data without removing old content.

```
with open("example.txt", "a") as file:  
    file.write("\nNew line added")
```

6. Using with Statement

Automatically closes the file, recommended best practice.

```
with open("example.txt", "r") as file:  
    content = file.read()
```

7. Exception Handling

Avoid crashes using try-except.

```
try:
    with open("file.txt", "r") as file:
        print(file.read())
except FileNotFoundError:
    print("File not found")
```

8. File Methods

Important methods: read(), readline(), readlines(), write().

9. CSV Files

CSV files are widely used for datasets in backend and ML.

```
import csv
with open("data.csv", "r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

10. JSON Files

JSON is used in APIs and data exchange.

```
import json
data = {"name": "Krishna", "age": 21}

with open("data.json", "w") as file:
    json.dump(data, file)

with open("data.json", "r") as file:
    data = json.load(file)
    print(data["name"])
```

11. Backend Use Cases

File handling is used for file uploads, logs, configs, and API data processing.

12. AI/ML Use Cases

Used for loading datasets, saving models, and storing predictions.

13. Practice Programs

```
# 1. Write file
with open("student.txt", "w") as file:
    file.write("Name: Krishna\nMarks: 85")

# 2. Read file
with open("student.txt", "r") as file:
    print(file.read())

# 3. Append file
with open("student.txt", "a") as file:
    file.write("\nGrade: A")

# 4. Word count
with open("student.txt", "r") as file:
    words = file.read().split()
print(len(words))

# 5. Copy file
with open("a.txt", "r") as src:
    data = src.read()

with open("b.txt", "w") as dest:
    dest.write(data)
```

14. Learning Order

1. open() 2. read/write 3. with statement 4. modes 5. exception handling 6. CSV 7. JSON